

# 2.1 Using and Manipulating Surds

Surd ⇒ a number written in form $\sqrt{a}$, $(a \neq n^2)$. Irrational

Simplification:

① $\sqrt{a}\sqrt{b} = \sqrt{ab}$

② $\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$

Rationalisation:

An equivalent form of a surd, but with an integer denominator

① $\frac{n}{\sqrt{a}} = \frac{n}{\sqrt{a}} \cdot \frac{\sqrt{a}}{\sqrt{a}} = \frac{n\sqrt{a}}{a}$

② $\frac{n}{b \pm \sqrt{a}} = \frac{n}{b \pm \sqrt{a}} \cdot \frac{b \mp \sqrt{a}}{b \mp \sqrt{a}} = \frac{n(b \mp \sqrt{a})}{b^2 - a}$

$\left(\text{as } (b \pm \sqrt{a})(b \mp \sqrt{a}) = b^2 \pm b\sqrt{a} \mp b\sqrt{a} - \sqrt{a^2}\right)$

# 2.2 Working with indices

An index is a power, e.g. $a^n$. $a$ is the BASE, $n$ is INDEX/POWER

① $a^m \cdot a^n = a^{m+n}$

② $\frac{a^m}{a^n} = a^{m-n}$

③ $(a^m)^n = a^{mn}$

④ $a^{-m} = \frac{1}{a^m}$

⑤ $a^{\frac{m}{n}} = \sqrt[n]{a^m} = \left(\sqrt[n]{a}\right)^m$

⑥ $a^0 = 1$

Different Bases: $(a \cdot b)^n = a^n \cdot b^n$

$216 = 6^3 = (2 \times 3)^3 = 2^3 \cdot 3^3$